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বিসিএস, ব্যাংক ও অন্যান্য সরকারি চাকুরির প্রস্তুতির

পূর্ণাঙ্গ লেকচার সিরিজ

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ইন্দ্রজিৎ চন্দ্র সরকার

৩৮তম বিসিএস

(সাধারণ শিক্ষা ক্যাডার)

প্রভাষক, গণিত বিভাগ

চিওড়া সরকারি কলেজ, কুমিল্লা।

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Course Process



Video Class দেখুন



Study করুন



Live Exam দিন

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আজকের টপিক

লেকচার - ১১

টপিক:

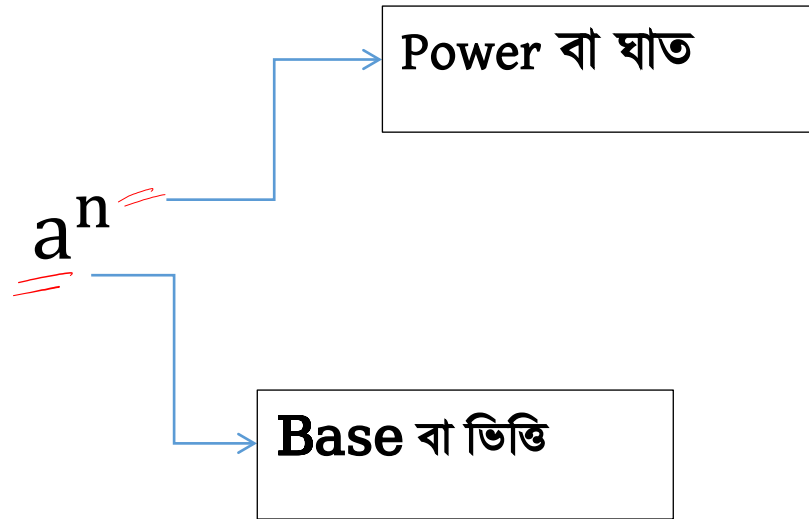
Exponential and Logarithms
Functions

(লগ ও সূচকীয় ফাংশন)



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সূচক



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বৈশিষ্ট্যঃ

$$\diamond a^m \times a^n = a^{m+n}, \quad a \neq 0$$

$$\diamond \frac{a^m}{a^n} = a^{m-n}$$

$$\diamond (a^n)^m = a^{mn}$$

$$\diamond \sqrt[n]{a} = a^{\frac{1}{n}}$$

$$\sqrt[n]{a} = a^{\frac{1}{n}}$$

$$\diamond (ab)^n = a^n b^n$$

$$\diamond \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\diamond a^0 = 1, a \neq 0 //$$

$$\diamond \text{যদি } a^x = b^x \text{ হয় তবে } a = b \text{ হবে।} //$$

$$\diamond \text{যদি } \underline{a^x} = \underline{a^y} \text{ হয় তবে, } x = y \text{ হবে।}$$

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০১। যদি $x^3\sqrt[3]{a} + y^3\sqrt[3]{b} + z^3\sqrt[3]{c} = 0$ এবং $a^2 = bc$ হয়, তবে দেখাও যে, $ax^3 + by^3 + cz^3 = 3axyz$

$$(\sqrt[3]{x})^3 = x$$

Solⁿ মডেল আছে,

$$x^3\sqrt[3]{a} + y^3\sqrt[3]{b} + z^3\sqrt[3]{c} = 0 \rightarrow \text{Trick 1}$$

$$\text{অ, } x^3\sqrt[3]{a} + y^3\sqrt[3]{b} = -z^3\sqrt[3]{c} \quad [\text{ফর জায়গায়ে}]$$

$$\text{অ, } (x^3\sqrt[3]{a} + y^3\sqrt[3]{b})^3 = (-z^3\sqrt[3]{c})^3$$

$$\text{অ, } (x^3\sqrt[3]{a})^3 + (y^3\sqrt[3]{b})^3 + 3x^3\sqrt[3]{a}y^3\sqrt[3]{b}(x^3\sqrt[3]{a} + y^3\sqrt[3]{b}) = -z^3c$$

$$\text{অ, } x^3a + y^3b + 3xyz\sqrt[3]{abc}(-z^3\sqrt[3]{c}) = -z^3c$$

$$\text{অ, } x^3a + y^3b - 3xyz\sqrt[3]{abc} = -z^3c$$

$$\text{অ, } ax^3 + by^3 - 3xyz\sqrt[3]{a \cdot a^2} = -z^3c \quad [\because bc = a^2]$$

$$\text{অ, } ax^3 + by^3 - 3xyz\sqrt[3]{a^3} = -z^3c$$

$$\text{অ, } ax^3 + by^3 - 3axyz = -cz^3$$

$$\therefore ax^3 + by^3 + cz^3 = 3axyz \quad [\text{প্রমাণিত}]$$

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নিজে করুন

যদি $a^2 + 2 = 3^{\frac{2}{3}} + 3^{-\frac{2}{3}}$ এবং $a \geq 0$ হয়, তবে দেখাও যে, $3a^3 + 9a = 8$

Solⁿ চরিত্র আছে,

$$a^2 + 2 = 3^{\frac{2}{3}} + 3^{-\frac{2}{3}}$$

$$\text{অ, } a^2 = (3^{\frac{1}{3}})^2 + (3^{-\frac{1}{3}})^2 - 2$$

$$= (3^{\frac{1}{3}})^2 + (3^{-\frac{1}{3}})^2 - 2 \cdot 3^{\frac{1}{3}} \cdot 3^{-\frac{1}{3}}$$

$$a^2 = (3^{\frac{1}{3}} - 3^{-\frac{1}{3}})^2$$

$$\therefore a = 3^{\frac{1}{3}} - 3^{-\frac{1}{3}} \quad [\text{অনুমান করে}] \text{ — ①}$$

$$\text{অ, } a^3 = (3^{\frac{1}{3}} - 3^{-\frac{1}{3}})^3 \quad [\text{ফল করে}]$$

$$\text{অ, } a^3 = (3^{\frac{1}{3}})^3 - (3^{-\frac{1}{3}})^3 - 3 \cdot 3^{\frac{1}{3}} \cdot 3^{-\frac{1}{3}} (3^{\frac{1}{3}} - 3^{-\frac{1}{3}})$$

$$\text{অ, } a^3 = 3 - 3^{-1} - 3 \cdot 3^{\frac{1}{3} - \frac{1}{3}} \cdot a \quad [\text{① এর মান আছে}]$$

$$\text{অ, } a^3 = 3 - \frac{1}{3} - 3a$$

$$[3^{\frac{1}{3}} \cdot 3^{-\frac{1}{3}} = 3^{\frac{1}{3} - \frac{1}{3}} = 3^0 = 1]$$



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$$a^3 = 3 - \frac{1}{3} - 3a$$
$$\Rightarrow a^3 = \frac{9 - 1 - 9a}{3}$$
$$\Rightarrow 3a^3 = 8 - 9a$$
$$\therefore 3a^3 + 9a - 8 = 0$$
$$\therefore 3a^3 + 9a = 8$$



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০২। সরল করঃ

$$\begin{aligned} & \frac{\left(\frac{a+b}{b}\right)^{\frac{a}{a-b}} \times \left(\frac{a-b}{a}\right)^{\frac{a}{a-b}}}{\left(\frac{a+b}{b}\right)^{\frac{b}{a-b}} \times \left(\frac{a-b}{a}\right)^{\frac{b}{a-b}}} \quad \text{base} \quad \text{base} \\ &= \left(\frac{a+b}{b}\right)^{\frac{a}{a-b} - \frac{b}{a-b}} \times \left(\frac{a-b}{a}\right)^{\frac{a}{a-b} - \frac{b}{a-b}} \\ &= \left(\frac{a+b}{b}\right)^{\frac{a-b}{a-b}} \times \left(\frac{a-b}{a}\right)^{\frac{a-b}{a-b}} \\ &= \frac{a+b}{b} \times \frac{a-b}{a} \\ &= \frac{a^2 - b^2}{ab} \end{aligned}$$



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নিজে করুন

$$a^{-m} = \frac{1}{a^m}$$

$$\frac{1}{1+a^{-m}b^n+a^{-m}c^p} + \frac{1}{1+b^{-n}c^p+b^{-n}a^m} + \frac{1}{1+c^{-p}a^m+c^{-p}b^n}$$

$$= \frac{1}{1 + \frac{b^n}{a^m} + \frac{c^p}{a^m}}$$

$$= \frac{1}{\frac{a^m + b^n + c^p}{a^m}}$$

$$= \frac{a^m}{a^m + b^n + c^p} + \frac{b^n}{a^m + b^n + c^p} + \frac{c^p}{a^m + b^n + c^p}$$

$$= \frac{a^m + b^n + c^p}{a^m + b^n + c^p} = 1$$

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৩। সমাধান করুন, $5^x + 3^y = 8$, $5^{x-1} + 3^{y-1} = 2$

Solⁿ দেয়া আছে, $5^x + 3^y = 8$
 $\therefore 3^y = 8 - 5^x$ — ①

আবার, $5^{x-1} + 3^{y-1} = 2$

অ, $\frac{5^x}{5} + \frac{3^y}{3} = 2$

অ, $\frac{5^x}{5} + \frac{8 - 5^x}{3} = 2$ [①-র ব্যবহার করা হয়েছে]

অ, $\frac{3 \cdot 5^x + 5(8 - 5^x)}{15} = 2$

অ, $3 \cdot 5^x + 40 - 5 \cdot 5^x = 2 \times 15$

অ, $40 - 2 \cdot 5^x = 30$

অ, $40 - 30 = 2 \cdot 5^x$

অ, $10 = 2 \cdot 5^x$

অ $5^x = \frac{10}{2} = 5$
অ $5^x = 5^1$
 $\therefore x = 1$



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① જ્યારે $x = 1$ સત્યતા મળે,

$$3^y = 8 - 5^1$$

$$\text{અ, } 3^y = 8 - 5$$

$$\text{અ, } 3^y = 3^1$$

$$\therefore y = 1$$

$\therefore$ નિશ્ચય સંભવિત $(x, y) = (1, 1)$



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নিজে করুন

✍ সমাধান করুন, $4^{3y-2} = 16^{x+y}$, $3^{x+2y} = 9^{2x+1}$

Solⁿ

$$4^{3y-2} = 16^{x+y}$$
$$\text{or } 4^{3y-2} = (4^2)^{x+y}$$
$$\text{or } 4^{(3y-2)} = 4^{(2x+2y)}$$
$$\therefore 3y-2 = 2x+2y$$

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$$8 \mid 8y^x - y^{2x} = 16 \text{ এবং } 2^x = y^2$$

Solⁿ অনুসরণ করুন

$$8y^x - y^{2x} = 16$$

$$\text{বা, } 8y^x - (y^x)^2 = 16$$

$$\text{বা, } 8a - a^2 = 16 \quad [\text{কি, } y^x = a]$$

$$\text{বা, } 0 = a^2 - 8a + 16$$

$$\text{বা, } 0 = a^2 - 2 \cdot 4 \cdot a + 4^2$$

$$\text{বা, } 0 = (a - 4)^2$$

$$\text{বা, } a - 4 = 0 \quad [\text{কর্মসূচ্য করুন}]$$

$$\text{বা, } a = 4$$

$$\text{বা, } y^x = 2^2 \text{ ——— ①}$$

$$\text{অতএব, } 2^x = y^2$$
$$\text{বা, } (2^x)^x = (y^2)^x$$

$$\text{বা, } 2^{x^2} = (y^x)^2$$

$$\text{বা, } 2^{x^2} = (2^2)^2$$

$$\text{বা, } 2^{x^2} = 2^4$$

$$\text{বা, } x^2 = 4$$

$$\therefore x = \pm 2$$



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ଏହା, $x = 2$ ହେଉଛି ① ନି ଚାହୁଁ,

$$y^2 = 2^2 \therefore y = \pm 2$$

ତାହା, $x = -2$ ହେଉଛି ① ନି ଚାହୁଁ,

$$y^{-2} = 2^2$$

$$\text{ଏ, } \frac{1}{y^2} = 2^2$$

$$\text{ଏ, } y^2 = \frac{1}{2^2}$$

$$\therefore y = \pm \frac{1}{2}$$

$$\therefore \text{ନିମ୍ନ ସମସ୍ତ (x, y) = (2, 2), (2, -2), (-2, \frac{1}{2}), (-2, -\frac{1}{2})$$



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নিজে করুন

$$18y^x - y^{2x} = 81 \text{ এবং } 3^x = y^2$$



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লগারিদম

$a^x = N$, ($a > 0, a \neq 1$) হলে, $x = \log_a N$ কে N এর a ভিত্তিক লগ বলা হয়।

দ্রষ্টব্য : x ধনাত্মক বা ঋণাত্মক যাই হোক না কেন, $a > 0$ হলে a^x সর্বদা ধনাত্মক। তাই শুধু ধনাত্মক সংখ্যারই লগের মান আছে যা বাস্তব। শূন্য বা ঋণাত্মক সংখ্যার লগের বাস্তব মান নেই।

বৈশিষ্ট্যঃ

- $\log_a M + \log_a N = \log_a (MN)$
- $\log_a M - \log_a N = \log_a \left(\frac{M}{N}\right)$
- $\log_a M^n = n \log_a M$
- $\log_a 1 = 0$
- $\log_a a^n = n$
- $\log_a Y = x$ হলে, $a^x = Y$
- $\log_a b = \frac{1}{\log_b a}$ $\equiv$
- বিঃদ্রঃ (log এর বেস বা ভিত্তি e হলে তা $\ln$ হবে) অর্থাৎ $\log_e X = \ln X$



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১. $\log_k \underline{ab} \log_k \left(\frac{a}{b}\right) + \log_k bc \log_k \left(\frac{b}{c}\right) + \log_k ca \log_k \left(\frac{c}{a}\right) = 0$ প্রমাণ করুন।

$$= (\log_k a + \log_k b) (\log_k a - \log_k b) + (\log_k b + \log_k c) (\log_k b - \log_k c) + (\log_k c + \log_k a) (\log_k c - \log_k a)$$

$$= (\cancel{\log_k a}^2 - \cancel{\log_k b}^2) + (\cancel{\log_k b}^2 - \cancel{\log_k c}^2) + (\cancel{\log_k c}^2 - \cancel{\log_k a}^2)$$

$$= 0$$

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$$\log\left(\frac{x+y}{3}\right) = \frac{1}{2}(\log x + \log y) \text{ প্রমাণ করুন, } \frac{x}{y} + \frac{y}{x} = 7$$

$$\log \frac{x+y}{3} = \frac{1}{2} \log(xy)$$

$$\therefore 2 \log \frac{x+y}{3} = \log xy$$

$$\therefore \log \left(\frac{x+y}{3}\right)^2 = \log xy$$

$$\therefore \frac{(x+y)^2}{9} = xy$$

$$\therefore \frac{(x+y)^2}{xy} = 9$$

$$\therefore \frac{x^2 + 2xy + y^2}{xy} = 9$$

$$\therefore \frac{x^2}{xy} + \frac{2xy}{xy} + \frac{y^2}{xy} = 9$$

$$\frac{x}{y} + 2 + \frac{y}{x} = 9$$

$$\therefore \frac{x}{y} + \frac{y}{x} = 9 - 2 = 7$$

(প্রমাণিত)



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২. যদি, $\frac{x(y+z-x)}{\log_k x} = \frac{y(z+x-y)}{\log_k y} = \frac{z(x+y-z)}{\log_k z}$ হয় তবে প্রমাণ কর যে,
 ~~$x^y y^x = y^z z^y = z^x x^z$~~ $x^y y^x = y^z z^y = x^z z^x$

Solⁿ $\frac{x(y+z-x)}{\log_k x} = \frac{y(z+x-y)}{\log_k y} = \frac{z(x+y-z)}{\log_k z} = p$ [ধরি]

অতঃ $\frac{x(y+z-x)}{\log_k x} = p \therefore \frac{x(y+z-x)}{p} = \log_k x$ — ①

এবং $\frac{y(z+x-y)}{\log_k y} = p \therefore \frac{y(z+x-y)}{p} = \log_k y$ — ②

এবং $\frac{z(x+y-z)}{\log_k z} = p \therefore \frac{z(x+y-z)}{p} = \log_k z$ — ③



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① $x \times y + ② \times x$ ਤਰ੍ਹਾਂ ਹੋਵੇ,

$$\frac{xy(y+z-x)}{p} + \frac{xy(x+z-y)}{p} = y \log_k x + x \log_k y$$

$$\text{ਅ, } \frac{xy(y+z-x+x+z-y)}{p} = \log_k x^y + \log_k y^x$$

$$\text{ੴ, } \frac{2xyz}{p} = \log_k x^y \cdot y^x$$

② $x \times z + ③ \times y$ ਤਰ੍ਹਾਂ ਹੋਵੇ,

$$\frac{zy(x+z-y)}{p} + \frac{zy(x+y-z)}{p} = z \log_k y + y \log_k z$$

$$\text{ਅ, } \frac{zy(x+z-y+x+y-z)}{p} = \log_k y^z + \log_k z^y$$

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$$\therefore \frac{2xyz}{p} = \log_k y^z z^y$$

ଅନୁସାରେ, ① $\times z$ + ③ $\times x$ ଟିଆଁ କରି,

$$\frac{2xyz}{p} = \log_k x^z z^x$$

$$\therefore \log_k x^y y^x = \log_k y^z z^y = \log_k z^x x^z$$

$$\therefore x^y y^x = y^z z^y = z^x x^z$$

[ସମୀକରଣ]



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নিজে করুন

$$\frac{ab \log_k ab}{\cancel{b+c} \atop a+b} = \frac{bc \log_k bc}{\cancel{c+a} \atop b+c} = \frac{ca \log_k ca}{\cancel{a+b} \atop c+a} \text{ হলে, প্রমাণ করুন, } a^a \cdot b^b \cdot c^c = 1$$
$$= p$$

$$ab \log_k ab = p(a+b)$$

$$\log_k ab = p \left(\frac{a+b}{ab} \right)$$

$$\text{বা, } \log_k a + \log_k b = p \left(\frac{1}{b} + \frac{1}{a} \right) \text{ --- (1)}$$

$$\text{অনুরূপভাবে, } \log_k b + \log_k c = p \left(\frac{1}{c} + \frac{1}{b} \right) \text{ --- (2)}$$

$$\text{বা, } \log_k c + \log_k a = p \left(\frac{1}{a} + \frac{1}{c} \right) \text{ --- (3)}$$

$$\text{(1) + (2) + (3) } \Rightarrow$$

$$\cancel{2} (\log_k a + \log_k b + \log_k c) = \cancel{2} p \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$

$$\therefore \log_k a + \log_k b + \log_k c = p \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \text{ --- (4)}$$

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$$\textcircled{4} \text{N} - \textcircled{2} \text{N}$$

$$\log_k a = p \times \frac{1}{a}$$

$$\text{u, } a \log_k a = p \times \frac{1}{a} \times a$$

$$\text{u, } \log_k a^a = p$$

$$\textcircled{4} \text{N} - \textcircled{1} \text{N} \Rightarrow \log_k c^c = p$$

$$\textcircled{4} \text{N} - \textcircled{3} \text{N} \Rightarrow \log_k b^b = p$$



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৩. যদি, $p = \log_a(bc)$, $q = \log_b(ca)$, $r = \log_c(ab)$ হয় তবে দেখাও যে, $\frac{1}{p+1} +$

$$\frac{1}{q+1} + \frac{1}{r+1} = 1$$

Solⁿ $p = \log_a bc$

$$\text{অ, } p+1 = \log_a bc + 1$$

$$= \log_a bc + \log_a a$$

$$= \log_a abc$$

$$\therefore \frac{1}{p+1} = \frac{1}{\log_a abc}$$

$$= \log_{abc} a$$

$$\left[\because \log_x y = \frac{1}{\log_y x} \right]$$

অনুরূপ, $q = \log_b ca$

$$\text{অ, } q+1 = \log_b ca + 1$$

$$= \log_b ca + \log_b b = \log_b abc$$



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$$\frac{1}{p+1} = \frac{1}{\log_b abc} = \log_{abc} b$$

অনুরূপ, $\frac{1}{r+1} = \log_{abc} c$

$$\begin{aligned}\therefore \frac{1}{p+1} + \frac{1}{q+1} + \frac{1}{r+1} &= \log_{abc} a + \log_{abc} b + \log_{abc} c \\ &= \log_{(abc)} (abc)^1 \\ &= 1\end{aligned}$$

[প্রমাণিত]



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নিজে করুন

$\frac{\log a}{b-c} = \frac{\log b}{z-x} = \frac{\log c}{x-y}$ হলে প্রমাণ করুন, $a^x b^y c^z = 1$

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০৪। $\log_2 y + \log_2(3y + 4) = 2 \log_2(3y - 4)$ সমাধান করুন। $y=4$

Solⁿ সমাধান করে,

$$\log_2 y + \log_2(3y + 4) = \underline{2} \log_2(3y - 4)$$

$$\text{অ, } \log_2 y(3y + 4) = \log_2 (3y - 4)^2$$

$$\text{অ, } y(3y + 4) = (3y - 4)^2$$

$$\text{অ, } 3y^2 + 4y = (3y)^2 - 2 \cdot 3y \cdot 4 + 4^2$$

$$\text{অ, } 3y^2 + 4y = 9y^2 - 24y + 16$$

$$\text{অ, } 0 = 9y^2 - 24y + 16 - 3y^2 - 4y$$

$$\text{অ, } 0 = 6y^2 - 28y + 16$$

$$\text{অ, } 0 = 2(3y^2 - 14y + 8)$$

$$\text{অ, } 0 = 3y^2 - 14y + 8$$



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$$3y^2 - 12y - 2y + 8 = 0$$

$$\text{or, } 3y(y-4) - 2(y-4) = 0$$

$$\text{or, } (y-4)(3y-2) = 0$$

$$\therefore \text{or } y-4 = 0$$

$$\therefore y = 4$$

$$\text{or } 3y-2 = 0$$

$$\therefore y = \frac{2}{3}$$

$$\text{or, } \log_2(3y-4), \quad x = \frac{2}{3} \text{ or } \text{or}$$

$$= \log_2\left(3 \cdot \frac{2}{3} - 4\right) = \log_2(-2)$$

$$\therefore x = \frac{2}{3} \text{ is not a solution}$$

$$\therefore \text{the only solution, } x = 4$$



Interactive Class

নিজে করুন

$$\begin{aligned} \log_{10}(x+4) + \log_{10}(x+16) &= 1 + 2 \log_{10} x \quad \boxed{x=4} \\ &= \boxed{\log_{10} 10} + \log_{10} x^2 \\ \log_{10} (x+4)(x+16) &= \log_{10} 10x^2 \end{aligned}$$



Live

Interactive Class

৪। $p = xy^{a-1}, q = xy^{b-1}, r = xy^{c-1}$ হলে প্রমাণ করুন, $\log p^{b-c} + \log q^{c-a} + \log r^{a-b} = 0$



Live

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Class**

নিজে করুন

যদি $x = 1 + \log_a(bc)$, $y = 1 + \log_b(ca)$, $z = 1 + \log_c(ab)$ হয়, তবে
প্রমাণ করুন যে, $xyz = xy + yz + zx$



Live

**Interactive
Class**

৫। যদি, $\frac{\ln a}{q-r} = \frac{\ln b}{r-p} = \frac{\ln c}{p-q}$ হলে, প্রমাণ করুন যে, $a^{q+r} \cdot b^{r+p} \cdot c^{p+q} = 1$



Live

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Class**

নিজে করুন,

$\frac{\log_k a}{b-c} = \frac{\log_k b}{c-a} = \frac{\log_k c}{a-b}$ হলে, প্রমাণ করুন, $a^a \cdot b^b \cdot c^c = 1$



Live
Interactive
Class



Live Interactive Class

ধন্যবাদ

ঘৰে বসে বিসিএস ও অন্যান্য চাকৰিৰ লিখিত প্ৰস্তুতি নিতে ▶



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